AQUAFAB INDUSTRIES

Getting Started with Aquafab Workbench

Download, install, activate a license, and find your way around.

For Aquafab Workbench 0.2.1

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Screenshots show simulated machines and modelled example data, not measurements. Aquafab Workbench is not a safety system.

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1 Getting started

Aquafab Workbench is the one application that runs every Aquafab flow chamber, respirometry system and swim tunnel, and analyses the data they produce. This guide takes you from download to your first (simulated) trial in about fifteen minutes. The full *Aquafab Workbench User Guide* covers every page in detail.

> Screenshots in this guide come from the real application running simulated machines. Numbers in them are simulated or modelled, not measurements.

1.1 Download

Go to **aquafabindustries.com/pages/aquafab-workbench** and open the *Downloads* section. No account is needed. Choose the installer for your computer:

Computer	Installer
Windows 10 or 11, 64-bit	`AquafabWorkbench-Windows.exe`
Mac with Apple silicon (M1 or later)	`AquafabWorkbench-macOS.dmg`
Mac with an Intel processor	`AquafabWorkbench-macOS-Intel.dmg`
Linux, x86-64	`AquafabWorkbench-Linux.AppImage`

The Raspberry Pi control computer inside an Aquafab system comes with the software already installed.

1.2 Install

Windows. Run the installer. If Windows shows *"Windows protected your PC"*, click **More info**, then **Run anyway** (the installer is not yet code-signed). It installs for your user only and needs no administrator rights. Open **Aquafab Workbench** from the Start menu.

Mac. Open the `.dmg` and drag **Aquafab Workbench** into **Applications**. The first time you open it, macOS may refuse; click **Done**, then go to **System Settings → Privacy & Security**, click **Open Anyway** and confirm. After that it opens normally. (Apple menu → About This Mac tells you whether you have Apple silicon or Intel.)

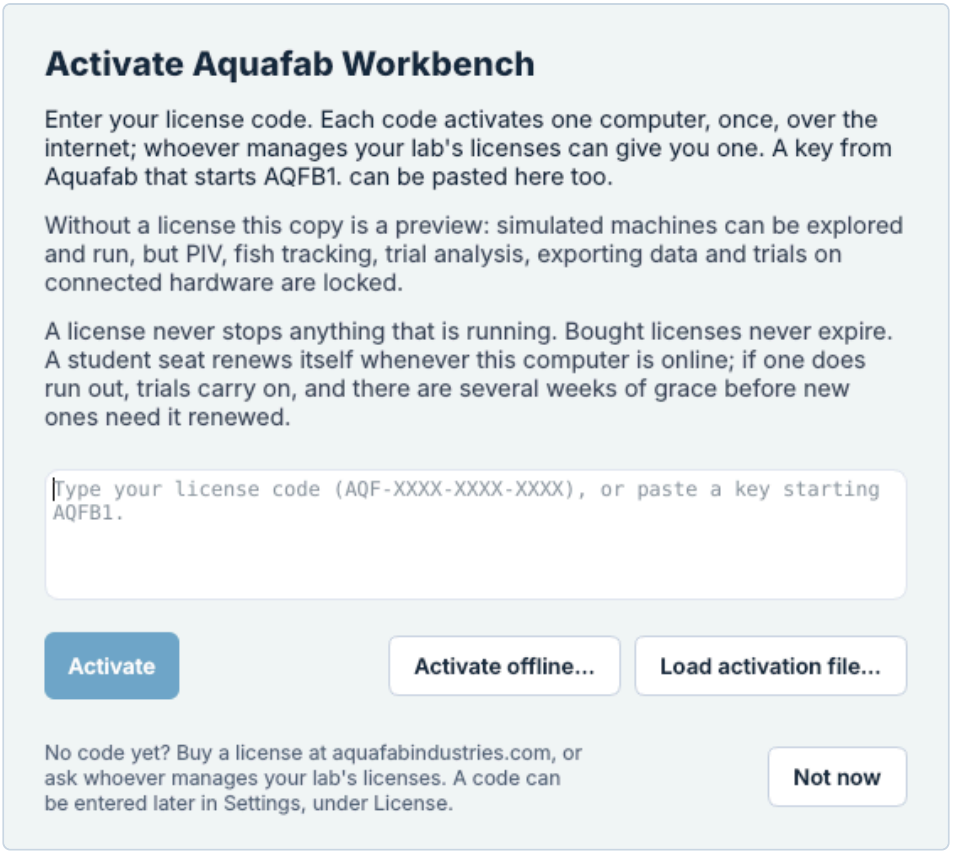
Linux. Make the AppImage executable and run it:

```
chmod +x AquafabWorkbench-Linux.AppImage
./AquafabWorkbench-Linux.AppImage
```

Raspberry Pi. It opens by itself, full screen, at power-on. Otherwise double-click the **Aquafab Workbench** icon on the desktop. **Exit full screen** in the top strip gives you an ordinary window; **Enter full screen** goes back.

1.3 Enter your license

The first time it opens, Aquafab Workbench asks for a license.

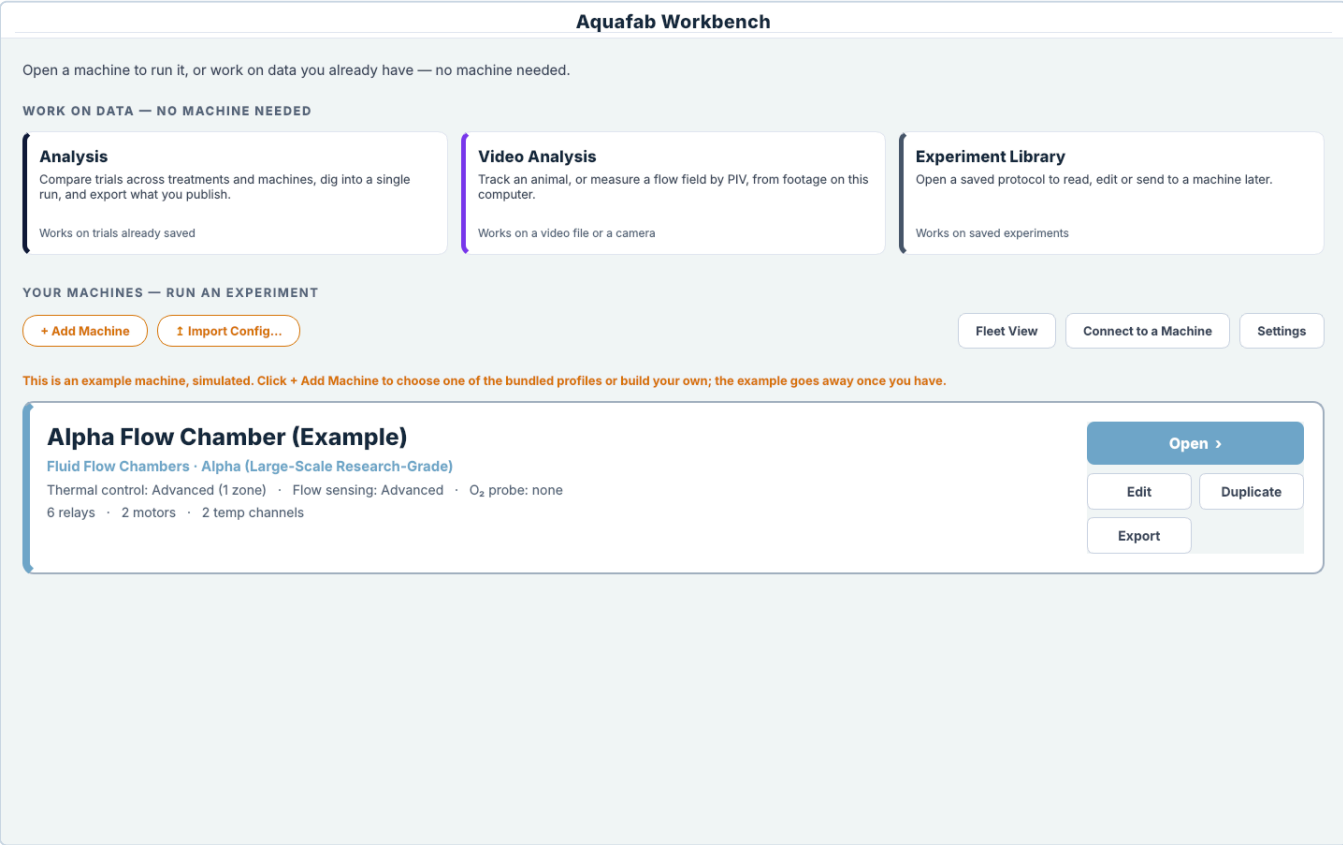


The license dialog.

- **If you have a code** (it looks like `AQF-XXXX-XXXX-XXXX`, from your lab's license manager or the confirmation of your purchase), type it and press **Activate**. The computer needs the internet for this, once.
- **If the computer has no internet**, type the code and press **Activate offline...**. It gives a request code; on any computer with internet, open **aquafabindustries.com/apps/licenses**, choose *Computer with no internet*, paste the request code and press *Make activation file*. Bring the file `aquafab-activation.txt` back and press **Load activation file...**.
- **If you do not have one yet**, press **Not now**. The software runs as a **preview**: the example machine works fully in simulation, and adding machines, PIV, fish tracking, analysis, export and trials on your own hardware show a *needs a license* card with **Buy a license** and **Enter a license code...**.

You can enter a license later under **Settings → License**. Each code activates one computer.

1.4 The Launcher



The Launcher.

The Launcher is the start screen; **‹ Home** in the top strip always returns to it.

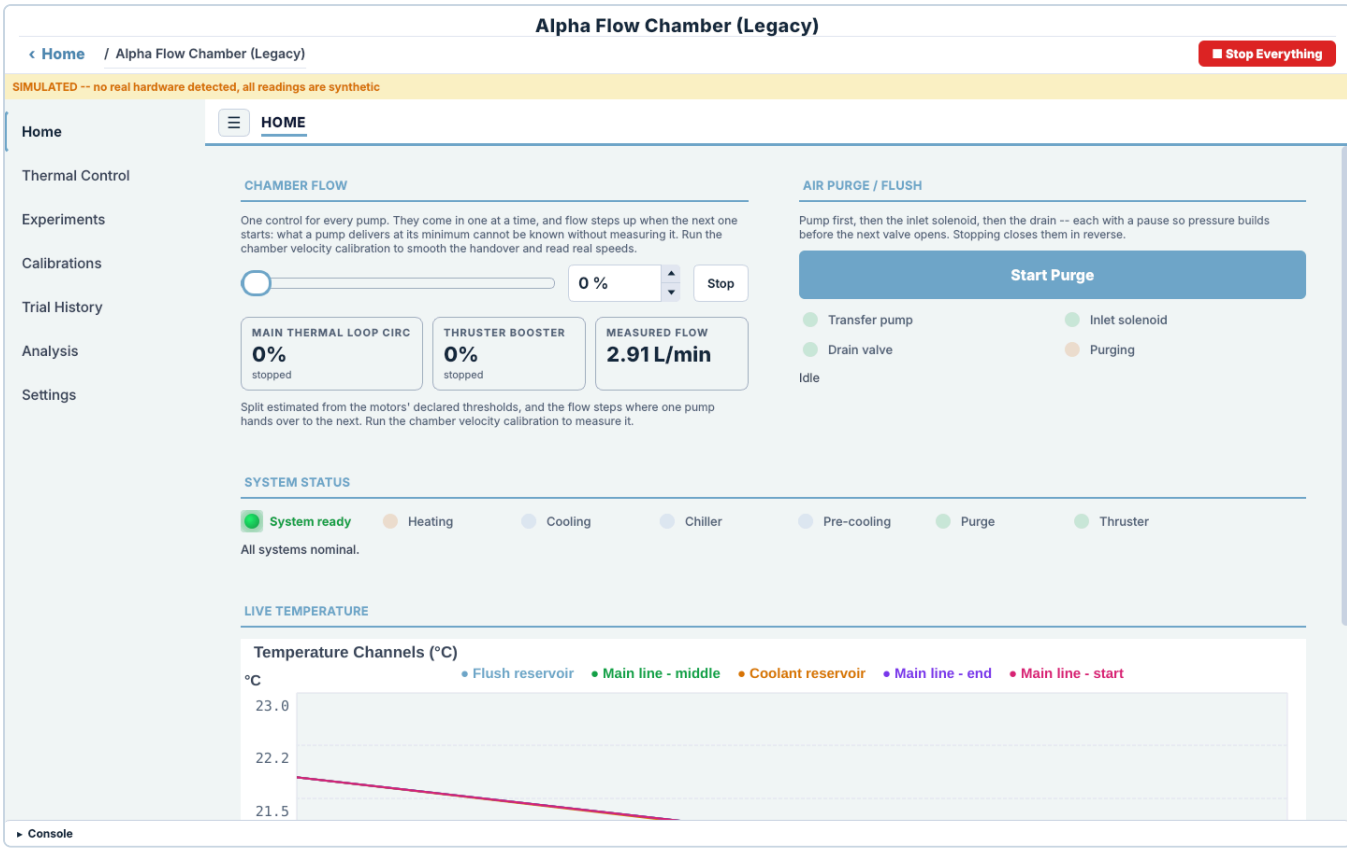
- **Work on data — no machine needed**: **Analysis** (every saved trial), **Video Analysis** (fish tracking and PIV on video files) and **Experiment Library** (saved protocols).
- **Your machines — run an experiment**: one row per machine, each with **Open ›**, **Edit**, **Duplicate**, **Export** and **Delete**. A new installation lists only **Alpha Flow Chamber (Example)**, always simulated.
- Above the list: **+ Add Machine**, **↥ Import Config...**, **Fleet View**, **Connect to a Machine** (reach a machine elsewhere, chapter 21 of the user guide) and **Settings**.

1.5 Add your machine

Press **+ Add Machine** (this needs a license). Choose your machine from the bundled profiles and press **Add**, or press **Build your own...** for a rig of your own. The example machine goes away once you have added one.

1.6 Open a machine

Press **Open ›** on a machine. On any computer that is not wired to that machine, it opens in **simulation**: a dialog titled *"Running in simulation"* explains why (the example machine, and any machine in the preview, skips the dialog), and an amber **SIMULATED** banner stays across the top. That is expected on a laptop. Simulation runs the whole machine against a built-in model, so you can learn the software with nothing at risk.



A machine's Home page, simulated.

The machine opens on **Home**. Down the left is the **section rail**:

Section	What it is for
Home	Live readings, flow and manual relay control.
Thermal Control	Holding a chamber at a temperature.
Experiments	Respirometry, temperature programs, swim tests, fish tracking, PIV, saved experiments.
Calibrations	Commissioning, chamber velocity, thermal step test, oxygen probe.
Trial History	Every saved trial on this machine.
Analysis	Comparing and re-analysing this machine's trials.
Settings	This machine's limits, emergency stop, remote access, console.

Hover over any control for a tooltip explaining it. ☰ hides the rail on a small screen.

■ **Stop Everything** at the top right (keyboard **Ctrl+.**, **⌘+.** on a Mac) stops every open machine: trials, pumps, heaters and relays. It asks first, and ⟲ **Undo Stop** puts things back. It is a convenience, not a substitute for the machine's physical emergency stop.

1.7 Try a trial on the simulator

1. Add and open the **Small Respirometry System**. (The example machine has no oxygen probe, so in the preview try a temperature program or a swim test on it instead.)
2. Go to **Experiments → Respirometry → Set up**.
3. Under **What are you measuring?**, choose **Respirometry**, then **Measurement: Standard metabolic rate (SMR)**.
4. Under **Chambers**, tick the chamber and type an **Animal mass (g)**.
5. Under **Rate analysis**, set **Simulation speed** to 300×, so the run takes minutes rather than hours.
6. Switch to **Live**, press **Prepare system**, then **Start Trial** (or **Start anyway**).
7. Watch the oxygen trace fall and recover with each cycle. Press **Stop** when you have seen enough. The trial has been saving itself after every cycle.
8. Open **Trial History** to see it saved, and **Inspect** it.

The *Running a Respirometry Trial* guide explains each step properly.

1.8 Keep it up to date

Aquafab Workbench checks for a new version shortly after it opens. When there is one, **Update available** appears in the top strip. Press it, then **Download and install**. Your machines, calibrations, trials and license are kept. An update is never installed while a trial is running.

1.9 Where next

To…	Read
Set up a new machine	*Commissioning a New Machine*
Measure metabolic rate	*Running a Respirometry Trial*
Run a tidal cycle or ramp	*Running a Temperature Program*
Measure flow, or calibrate pump output to cm/s	*PIV Flow Measurement and Chamber Velocity Calibration*
Measure U_{crit}	*Swim Performance (Ucrit) Testing*
Track an animal in video	*Fish Tracking*
Compare treatments and export	*Analysing and Exporting Results*
Manage codes and seats	*Managing Your Lab's Licenses*
Anything else	*Aquafab Workbench User Guide*

Help: **sales@aquafabindustries.com**, **+1 (513) 781-8010**.